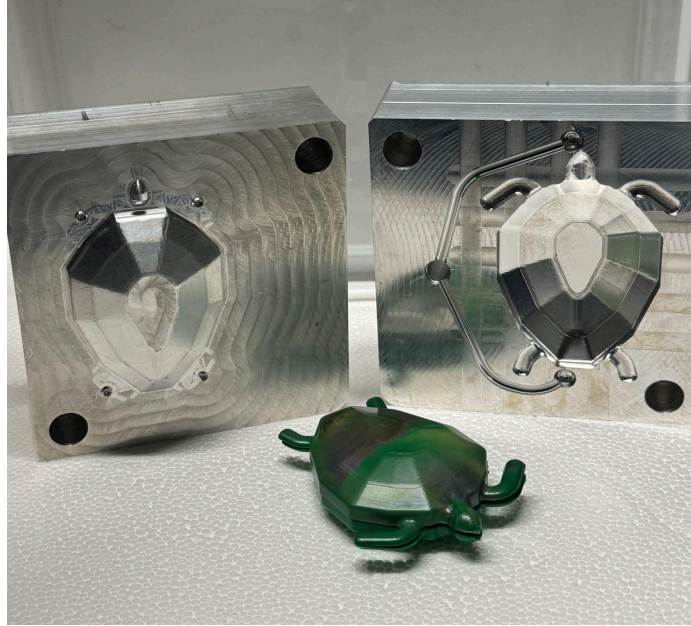




ME 340-2 Final Project: Turtle

Tomasz Krzeminski, Jeremy Lu, Gabriela Norwood, Natalia Yepes

Table of Contents

I. Introduction.....	3
II. Part Design.....	4
III. Mold Design.....	7
IV. Manufacturing Process.....	8
V. Molding Process.....	18
VI. Metrology Study.....	20
VII. Conclusion.....	26
VIII. Appendices.....	27

I. Introduction

The goal of this project was to design and manufacture an injection-molded plastic part from concept to production. This involved designing the part in CAD, mold design, CNC machining of the mold halves, and the final injection molding of the plastic components. This project provided hands-on experience with DFM and DFA concepts and navigating the constraints and design considerations for both CNC machining and injection molding.

For our project, we designed and manufactured a small plastic turtle. The turtle was selected as our part concept because its geometry allowed us to incorporate multiple features, such as the shell, head, and legs, while still allowing the design to be manufactured within the constraints of the three-axis CNC mill and injection molding process. The part was designed as two halves (top and bottom of the turtle) that would be molded separately and later assembled to form the complete turtle.

This report documents the development process of the turtle part, beginning with the initial concept and CAD design, followed by mold design and machining. It then describes the injection molding process used to produce the final parts and concludes with reflections on the challenges encountered and lessons learned throughout the project.

Figure 1 below shows the team's initial sketches and design for our turtle part, featuring four orthographic views of the part. The intent was to create two halves, a top and bottom, that align using both press and snap fit mating features.

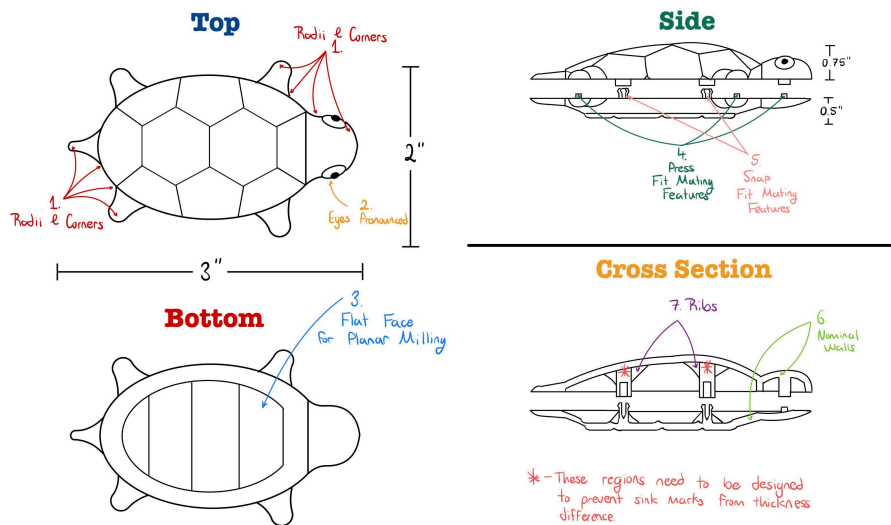


Figure 1. Initial sketches of our turtle part

Figure 2 showcases the CAD models of the final turtle, showcasing the geometry and general features of the top and bottom parts of the turtle. The left model is the top of the turtle, while the

right model in Figure 2 is the bottom of the turtle. The primary difference between the two halves is that the top of the turtle has an extra layer, and its head is a little longer than the bottom. The exact dimensions of the CAD models can be seen in Appendix A.

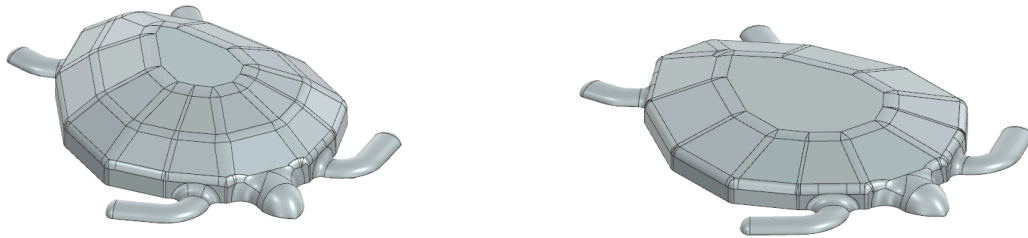


Figure 2. CAD models of top and bottom half of our turtle

II. Part Design

Many aspects of the design changed between the initial sketches shown in Figure 1 and the final part. The head was reduced in size, the tail feature was removed, ribs were not included as they were deemed unnecessary, and the final assembly only used press-fit connections instead of both press and snap fits. Despite these modifications, the original sketches served as a valuable starting point for developing the first CAD models and provided a visual reference for how the different components of the turtle would align during assembly.

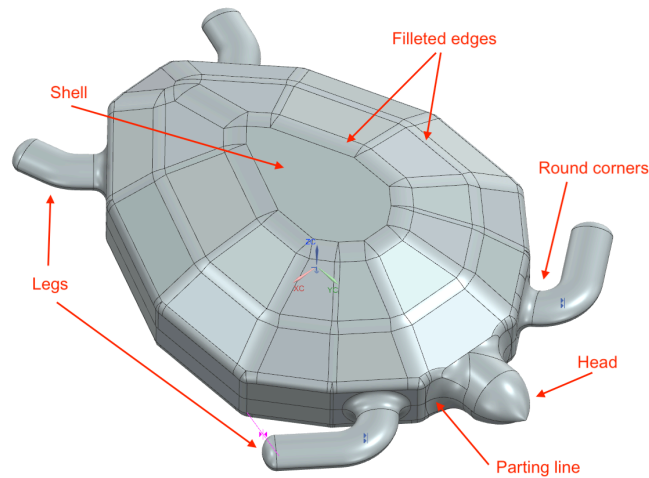


Figure 3. Isometric view of assembled turtle

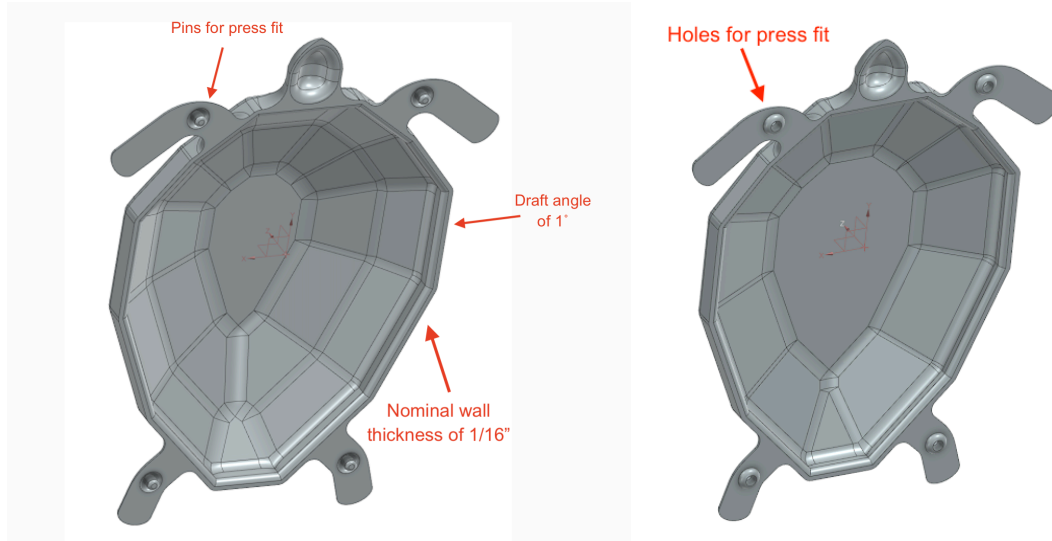


Figure 4. Top half of turtle (left) and bottom half of turtle (right)

Figures 3 and 4 show our final CAD design of the turtle, highlighting various features incorporated to improve machinability and quality of the injection molded part. A nominal wall thickness of 1/16" was maintained throughout the part to help minimize sink marks and reduce the risk of warpage during cooling. A draft angle of 1° was also applied to vertical faces to ensure proper ejection from the mold. Additionally, corners and edges were filleted to reduce stress concentrations and improve material flow during the injection molding process.

The design underwent several iterations as modifications were made to improve manufacturability. One notable change was the removal of the turtle's eyes after determining that the resulting overhang would be difficult to machine using a three-axis CNC mill. The tail was also removed because its small size would have added unnecessary complexity to the mold without providing significant value to the final part. Additionally, initial iterations had a more complex shell structure with more faces and an undefined layer height. This ultimately resulted in increased machining time and complexity as smaller tools would have had to be used to machine between each face. Other minor modifications included adding fillets and external radii to sharp corners for improving machinability.

Figure 4 shows the pin-hole interference fits used to assemble the turtle. Early design sketches (Figure 1) included both snap fits and press fits as mating features, however, the snap fits were ultimately removed from the design because they would have significantly increased modeling complexity and required tighter control of feature geometry during machining and molding. Since press fits were sufficient to successfully assemble the turtle, the design was simplified to solely use press-fit connections.

To create these features, dowel pins were inserted into the bottom half of the mold during the molding process to form the hole features, while the corresponding pin features were included on the top half of the turtle. During machining, a #41 drill bit (0.096”) was used to create the pin features on the top half, while a 3/32” drill bit (0.0938”) was used to create the hole features on the bottom half. This difference in diameters produced the desired interference fit between the components. In order to determine the press fit calculations, equations (1) and (2) were used.

$$I = \left[\frac{(S_d \times d)}{W} \right] \times \left[\left(\frac{W + v_h}{E_h} \right) + \left(\frac{1 - v_s}{E_s} \right) \right] \quad (1)$$

$$I = d - d_1 \quad (2)$$

$$W = \frac{1 + \frac{d}{D}}{1 - \frac{d}{D}} \quad (3)$$

Symbol	Term	Value
S_d	Design Stress	30 MPa
D	Outside Hub Diameter	0.125 in
d	Pin Diameter	0.09556 in
E_h	Hub Young's Modulus	2000 MPa
E_s	Shaft Young's Modulus	2000 MPa
v_h	Hub Poisson's Ratio	0.435
v_s	Shaft Poisson's Ratio	0.435

$$W = \frac{1 + \frac{0.09556}{0.125}}{1 - \frac{0.09556}{0.125}} = 1.667 \quad (4)$$

$$I = \left[\frac{(30 \times 0.09556)}{1.667} \right] \times \left[\left(\frac{1.667 + 0.435}{2000} \right) + \left(\frac{1 - 0.435}{2000} \right) \right] = 0.001807 \quad (5)$$

$$d_1 = 0.09556 - 0.001807 = \mathbf{0.0938 \text{ in.}} \quad (6)$$

III. Mold Design

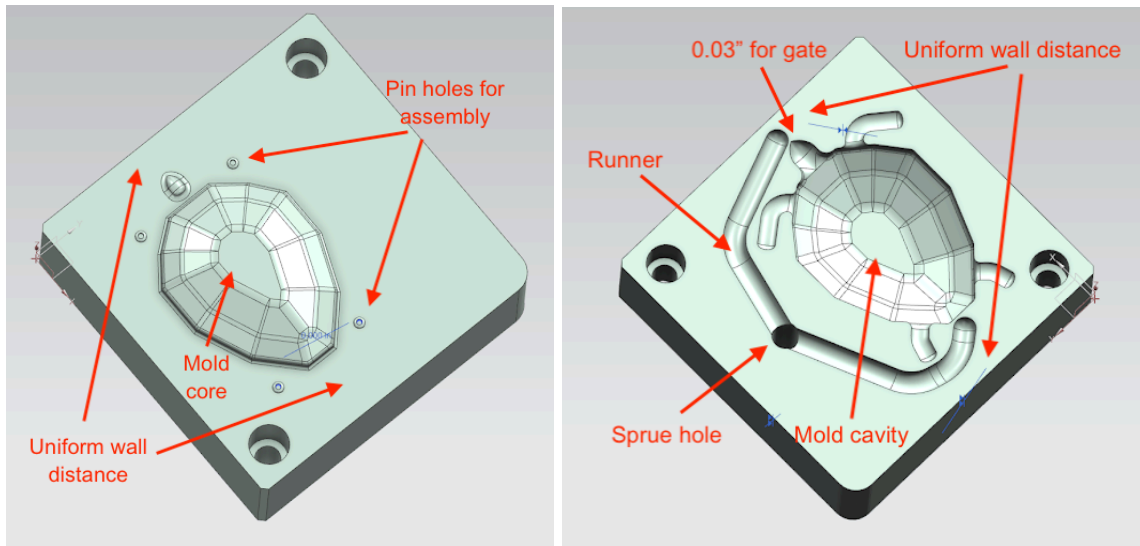


Figure 5. Core (left) and cavity (right) molds for top half of turtle

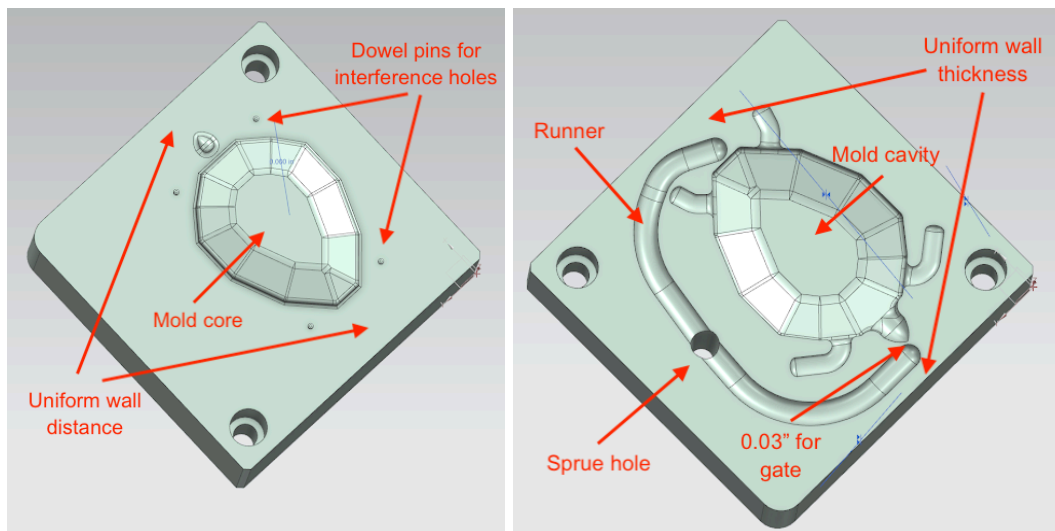


Figure 6. Core (left) and cavity (right) molds for bottom half of turtle

The parting line for the turtle was selected along the natural edge of the shell, which was the most symmetrical orientation and also allowed the geometry of each part to be separated cleanly while avoiding undercuts that would be difficult or impossible to machine. This placement also simplified the machining by ensuring that all features of each cavity could be accessed with standard end milling operations while still maintaining the intended geometry of the turtle.

Many additional manufacturability considerations were incorporated into the mold design. The part was oriented lengthwise relative to the mold blank so that it would fit within the available area between the sprue and the mold edge. This orientation ensured sufficient clearance between

the part geometry and the mold boundaries as well as a relatively uniform distance between the two to prevent flash.

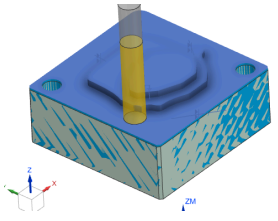
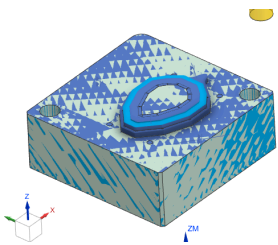
Shrinkage of the molded plastic was also considered during the design process. Dimensions of the cavity were slightly adjusted to account for expected material shrinkage so that the final molded part would more closely match with the intended design.

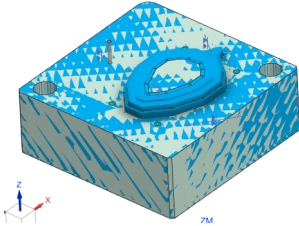
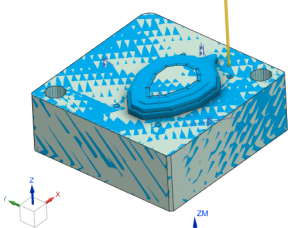
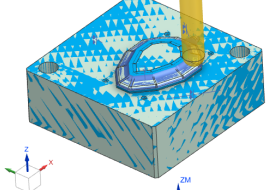
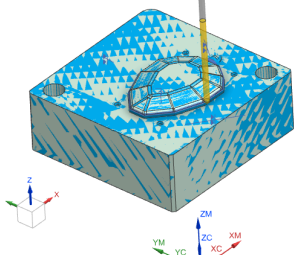
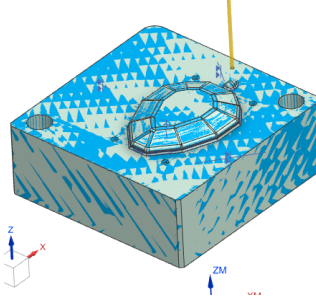
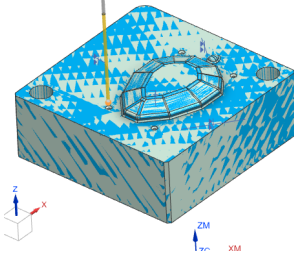
The runner system was designed with the opposing entrances into the part for both the top and bottom half of the turtle: one leading into the turtle's head and the other to the opposite end of the shell. This configuration allowed the molten plastic to enter the cavity from both directions along the central axis, helping to equalize the filling time and uniform material flow. A circular cross section was selected for the runner because it provides the most efficient flow characteristics and minimizes pressure losses compared to other runner geometries. The runner diameter was chosen to be 0.25", which is a standard size that provides sufficient material flow for parts of this scale. Additionally, the runner was placed 0.030" away from the part geometry to allow for manual filing of the gate after machining.

IV. Manufacturing Process

The following table outlines the CAM processes used to machine the molds. Separate core and cavity molds were created for both the top and bottom half of the turtle.

Table 1. Bottom core mold operations (2h 35min 34s)

Operation	Tool	Purpose	Preview
Adaptive Mill	EM_0.5	Roughing operation to remove a lot of material.	
Floor Wall	EM_0.5	Finishing operation to remove a lot of material from the floor of the mold and some material from the core part.	

Spot Drill	CENTERDRILL_N2	Initial center drilling operation for press fit features.	
Drill	STD_DRILL_3/32	Drilling operation for press fit features.	
Area Mill	BM_0.5	Finishing operation to remove material for curved features of turtle core.	
Area Mill	BM_0.125	Finishing operation to remove material around side walls and head of core.	
Area Mill	BM_0.0625	Finishing operation to remove material from around the side walls and head of the core.	
Planar Mill	BM_0.0625	Final finishing operation to remove material from press fit features.	

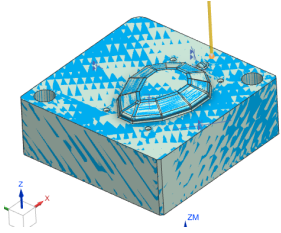
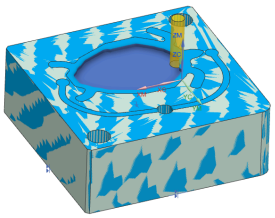
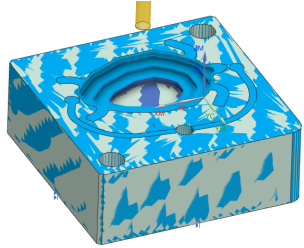
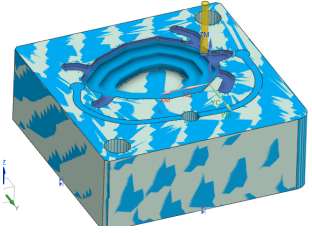
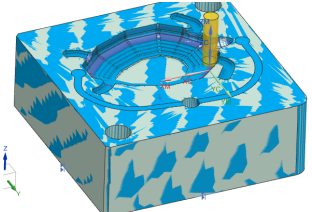
Floor Wall	EM_0.0625	Finishing operation to remove the last bit of material from around the head of the core.	
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Table 2. Bottom cavity mold operations (3h 21min 26s)

Operation	Tool	Purpose	Preview
Cavity Mill	EM_0.25	Roughing operation to remove initial bulk of material from blank.	
Floor Wall	EM_0.25	Finishing operation to remove material from flat bottom wall of turtle mold cavity.	
Cavity Mill	EM_0.125	Finishing operation to remove material from the entire cavity.	
Area Mill	BM_0.25	Finishing operation to remove material from curved inside surfaces of turtle cavity mold.	

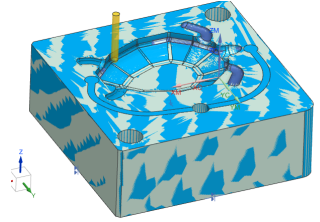
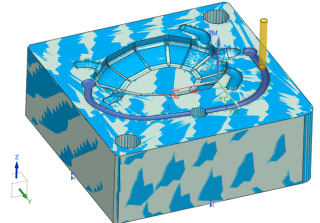
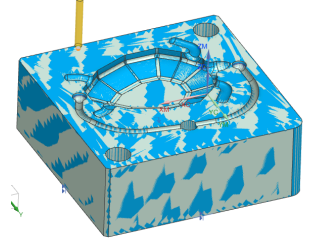
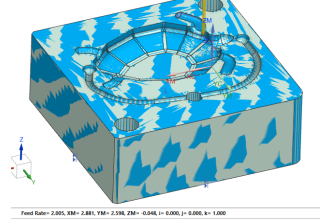
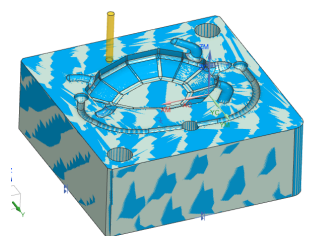
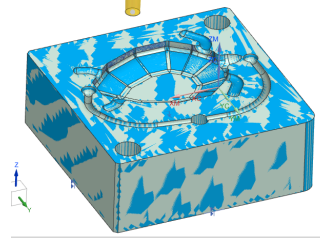
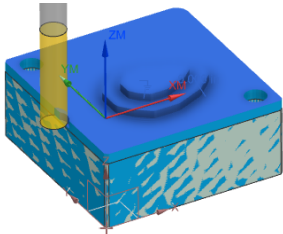
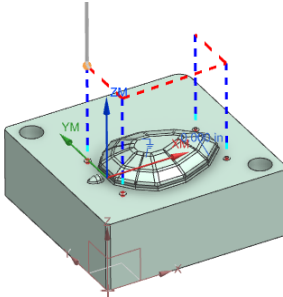
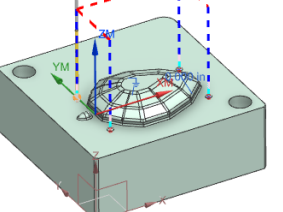
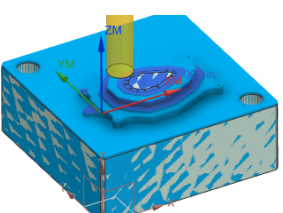
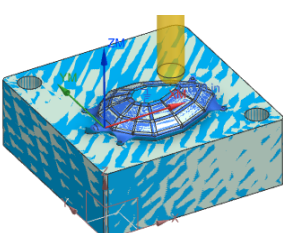
Area Mill	BM_0.125	Finishing operation to remove material from legs and upper edges of cavity mold.	
Planar Mill	BM_0.125	Finishing operation to remove material for runners.	
Area Mill	BM_0.125	Finishing operation to remove material from the top of the runners near where the gates will be placed.	
Area Mill	BM_0.0625	Finishing operation to remove material from leg and head areas of the turtle cavity mold and obtain a smoother surface finish.	 Feed Rate = 2.000, BM = 2.000, VM = 2.500, ZM = 0.000, L = 0.000, R = 1.000
Area Mill	BM_0.125	Final finishing operation to remove material from all areas of the turtle cavity mold.	
Planar Mill	TM_0.25	Final one-pass finishing operation to remove excess material on side walls.	

Table 3. Top core mold operations (2h 27min 50s)

Operation	Tool	Purpose	Preview
Adaptive Mill	EM_0.5	Roughing operation to remove a lot of material	
Spot Drill	SD_0.0625	Pre-drilling operation for press-fit features	
Drilling	DRILL_0.0625	Drilling press-fit features	
Floor Wall	EM_0.5	Finishing operation to remove details for initial part details and bulk of remaining material on floor wall	
Area Mill	BM_0.5	Finishing operation to remove material for flat and curved part details	

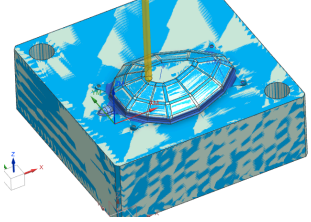
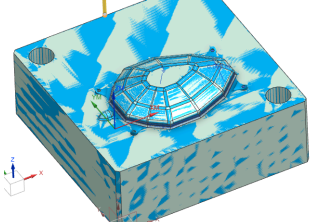
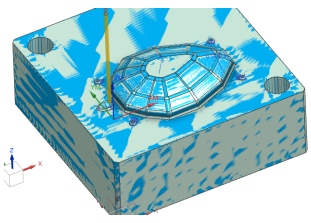
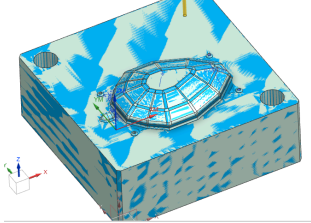
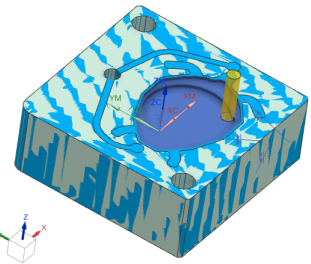
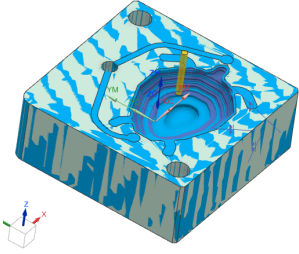
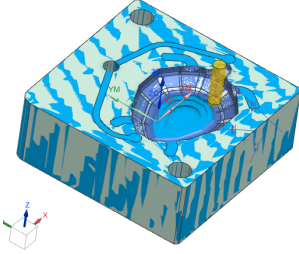
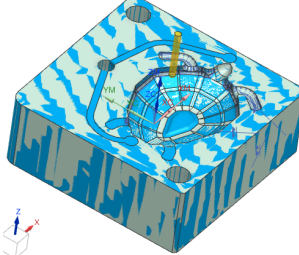
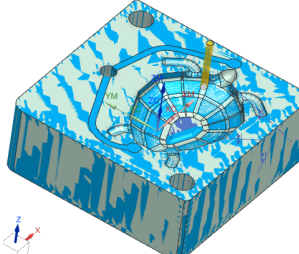
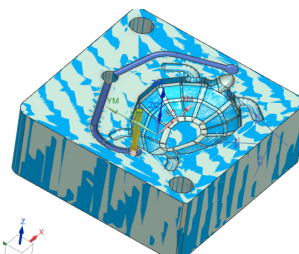
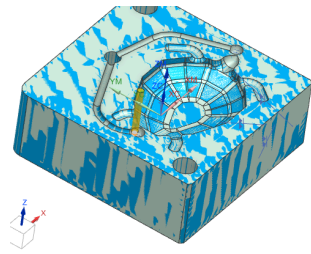
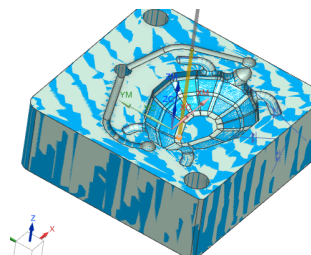
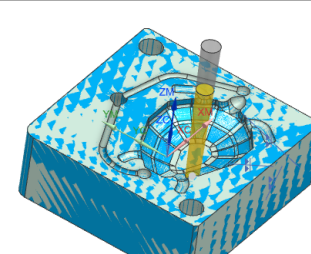
Area Mill	BM_0.125	Finishing operation to remove excess material near head, legs, and press fit areas.	
Area Mill	BM_0.0625	Finishing operation to remove more material near head, legs, and press fit area.	
Area Mill	BM_0.0625	Finishing operation to remove final remnants of material around the head and create the outer component of the press fit features.	
Floor Wall	EM_0.0625	Finishing operation to clean up the flat face of mold around the head, legs, and press fit areas.	

Table 4. Top cavity mold operations (2h 32min 52s)

Operation	Tool	Purpose	Preview
Adaptive Mill	EM_0.25	Roughing operation to remove initial material from the turtle body cavity.	

Adaptive Mill	EM_0.125	Roughing operation to remove more material from the cavity of the turtle body and head.	
Area Mill	BM_0.25	Finishing operation to remove material leftover from previous operations and begin to define features of the cavity body and head.	
Area Mill	BM_0.125	Finishing operation to remove excess material from the body cavity and get a smooth surface finish and remove initial material from legs.	
Floor Wall	EM_0.125	Finishing operation to remove material from the innermost flat surface that the previous tool was too large to reach.	
Planar Mill	BM_0.125	Finishing operation to remove material for the runner.	

Area Mill	BM_0.125	Finishing operation to remove material near the end of the runner/beginning of gates.	
Area Mill	BM_0.0625	Finishing operation to remove excess material from legs and clean up surface finish.	
Planar Mill	TM_0.25	Finishing operation to remove final leftover material and clean up surface finish.	

CAM Modifications

During our first machining attempt, we began by machining the core for the bottom half of the turtle. The original CAM used a #51 drill bit (0.0595") to create the holes for our mating features. However, during the drilling operation the drill bit broke due to its very small diameter and fragility during machining. To address this issue, we modified the design to accommodate a larger drill size. The hole diameter on the bottom core was increased by about 60% to a 3/32" drill bit (0.0938"). Correspondingly, the mating pin feature on the top core mold was upsized from 0.0625" to 0.096". After making these adjustments and updating the CAM operations, the machining process was completed successfully without further issues.

Additionally, chatter during machining negatively affected the geometry of the mating features. The design originally included very small lips around the holes for the dowel pins, but these features were damaged slightly during CNC machining due to their small size and lack of rigidity. As a result, the intended geometry of these features was not fully preserved. This ultimately impacted the quality of the assembly features during injection molding, which will be discussed in the following section.

Documentation of Manufacturing Process



Figure 7. One of our team members, Gabriela, adjusting the coolant on the CNC



Figure 8. One of our team members, Jeremy, finding collets for our tools



Figure 9. Disobeying shop rules and sharing snacks during an eight hour CNC shift

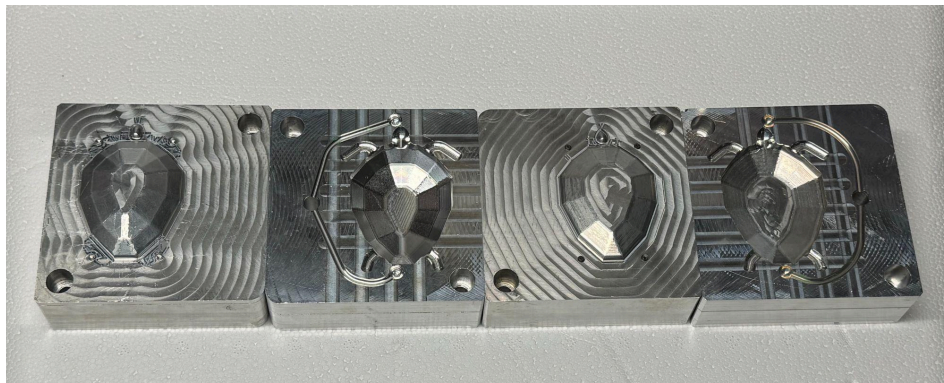


Figure 10. Finished molds: top half core and cavity (left) and bottom half core and cavity (right)

V. Molding Process



Figure 11. Our team inspecting an injection molded part



Figure 12. Assembled turtle (left), inside of disassembled parts (middle), outside of disassembled parts (right)

During the injection molding process, the barrel and nozzle temperature setpoints were both set to 450°F. While the barrel temperature reached its setpoint, the nozzle temperature consistently fluctuated around 400°F and did not fully reach the intended temperature. To compensate for this variation, injection times between 5s and 10s were used depending on the mold being run and the current nozzle temperature. The bottom half of the turtle required a shorter injection time because it had a smaller cavity volume. When the nozzle temperature was lower, longer injection

times were required to ensure the mold filled completely. A cooling time of 20 seconds was used before ejecting the part from the mold.

The molding process involved some trial and error. Initially, we selected an injection time that produced acceptable parts for the first few runs, but we soon began experiencing short shots and incomplete filling of the mold. Increasing the injection time alone did not resolve the issue. At first, we suspected the relatively low nozzle temperature was preventing proper material flow. However, after further troubleshooting we discovered that pellet buildup inside the pellet chamber was restricting the flow of molten plastic. Once this was identified, we found that cleaning the chamber every three to four injection cycles prevented the short shots from occurring.

We also encountered some difficulties during part ejection. The parts were removed by pulling on the legs of the turtle, which were relatively thin and delicate features. Because the plastic was still warm during ejection, pulling on the legs caused them to bend slightly upward. This can be seen in Figure 13 below. This deformation made assembly more difficult, particularly because the mating press-fit features were located on the legs.

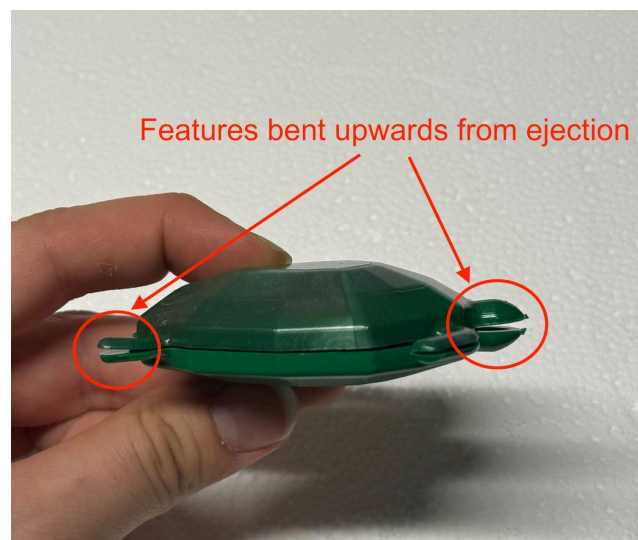


Figure 13. Gaps along parting line from bent features

Despite these challenges, the molding process overall produced good results. The molded turtles had good surface finish and did not show significant defects such as sink marks or warping. However, as mentioned in the previous section, chatter during CNC machining damaged some of the mating features in the mold. This affected the geometry of the press-fit features and made the final assembly less secure than originally intended.

VI. Metrology Study

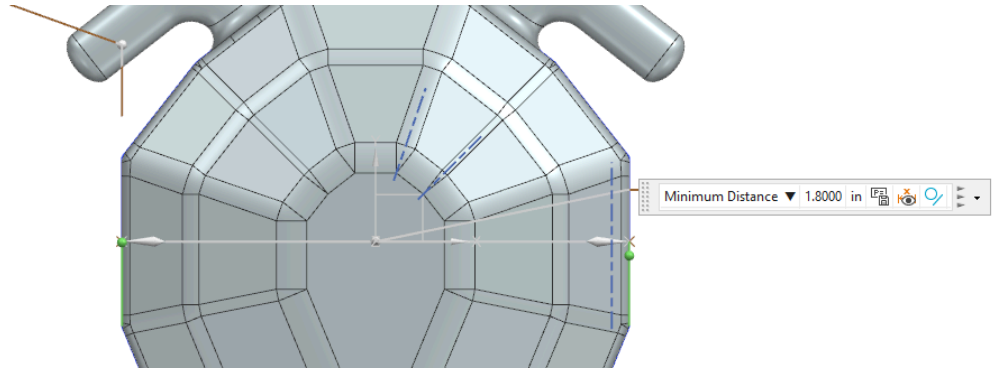


Figure 14. Selected geometry for metrology study: width of turtle

The selected geometry for the metrology study was the overall width of the turtle. The nominal dimension from the CAD model was 1.80in. Based on the expected variation in the turtle and the typical injection molding tolerances of about ± 0.010 in, we determined that a ± 0.010 in tolerance would be achievable for our small, thin-walled turtle. As a result of our tolerance specification the specification limit range was determined to be 1.810 - 1.790in for the top and bottom half of the turtle.



Figure 15. All 20 parts of the top half of the turtle

Table 5. Sample data for turtle top half width

	Sample						
Set	1	2	3	4	5	Average	Range
1	1.799	1.797	1.799	1.794	1.795	1.7968	0.005
2	1.789	1.789	1.785	1.795	1.796	1.7908	0.011
3	1.796	1.798	1.796	1.797	1.799	1.7972	0.003
4	1.796	1.797	1.797	1.795	1.796	1.7962	0.002
					Sum	7.181	0.021
					Average	1.79525	0.00525

$$UCL = \bar{\bar{x}} + A_2 \bar{R}$$

$$LCL = \bar{\bar{x}} - A_2 \bar{R}$$

Figure 16. Equations for UCL and LCL using sample data

Based on Table 5 above, which shows data for all four of our teammates taking five measurements each with a caliper, and using the equations in Figure 16 we can then calculate UCL and LCL values for our sample data as shown below:

$$A_2 = 0.577$$

$$UCL = 1.79525 + (0.577 * 0.00525) = 1.79828$$

$$LCL = 1.79525 - (0.577 * 0.00525) = 1.79222$$

Top Half Turtle Widths

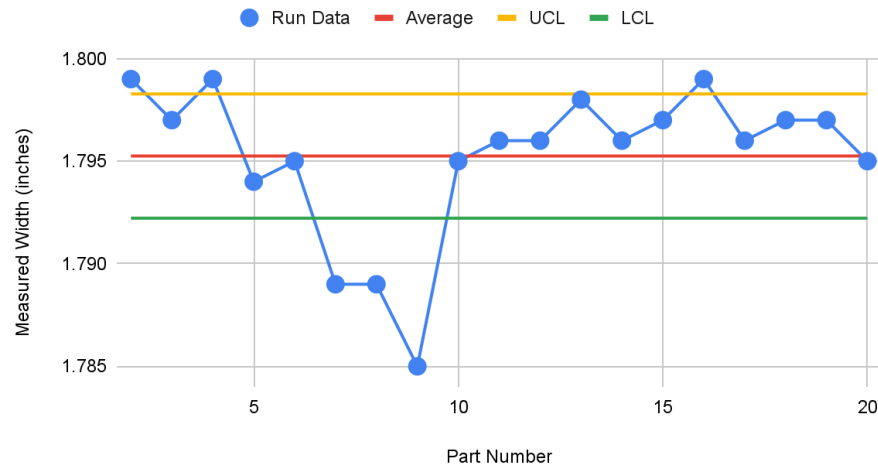


Figure 17. Run data of the top half widths

As seen in Figure 17 above, a run chart was created using measurements from the twenty top parts. While thirteen out of twenty parts fall within the control limits, we can see several trends that indicate the process is unstable. The first condition that makes it clear that this process is out of control is the fact that six parts fall outside of the control limits, satisfying condition 1. Condition 2 to determine an out of control process is also satisfied, with parts 7, 8, and 9 falling outside of the secondary range. Condition 5 is also met with nine consecutive points (points 11 - 19) all falling above the mean.

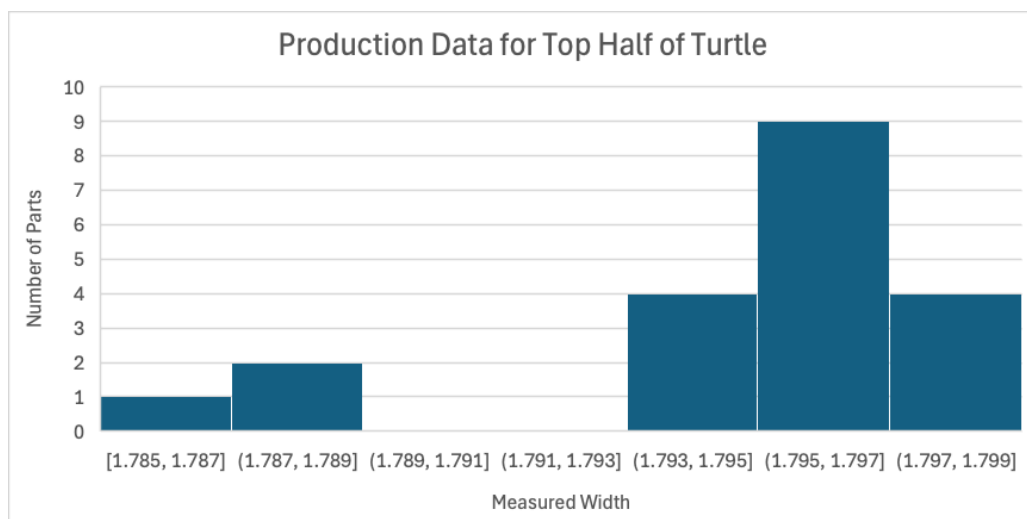


Figure 18. Histogram of measurement distribution of 20 top halves

From the histogram seen in Figure 18, it is clear that the parts do not have a normal distribution. Despite the instability of our process, only three parts fall outside of the specification limits. Further investigation into the process would be useful to determine exact causes of the out of control behavior seen here. Several factors could have contributed including inconsistency with the nozzle temperature and injection molding time, not enough consideration for shrinkage during the design process, and changes in operator as not all team members could always be present for injection molding.



Figure 19. All 20 parts of the bottom half of the turtle

We measured all 20 parts of the bottom half of the turtle with a caliper and calculated the average and UCL and LCL values. The USL and LSL values are, similarly, 1.810 and 1.790in.

Table 6. Sample data for turtle bottom half width

Set	Sample					Average	Range
	1	2	3	4	5		
1	1.792	1.777	1.778	1.779	1.775	1.7802	0.017
2	1.782	1.78	1.784	1.786	1.786	1.7836	0.006
3	1.787	1.784	1.785	1.784	1.789	1.7858	0.005
4	1.79	1.791	1.788	1.79	1.783	1.7884	0.008
					Sum	7.138	0.036
					Average	1.7845	0.009

Using the same process outlined above with the equations in Figure 16 and the sample data in Table 6, we can calculate the UCL and LCL for the bottom half of the turtle widths as shown below:

$$A_2 = 0.577$$

$$UCL = 1.7845 + (0.577 * 0.009) = 1.789693$$

$$LCL = 1.7845 - (0.577 * 0.009) = 1.779307$$

Bottom Half Turtle Widths

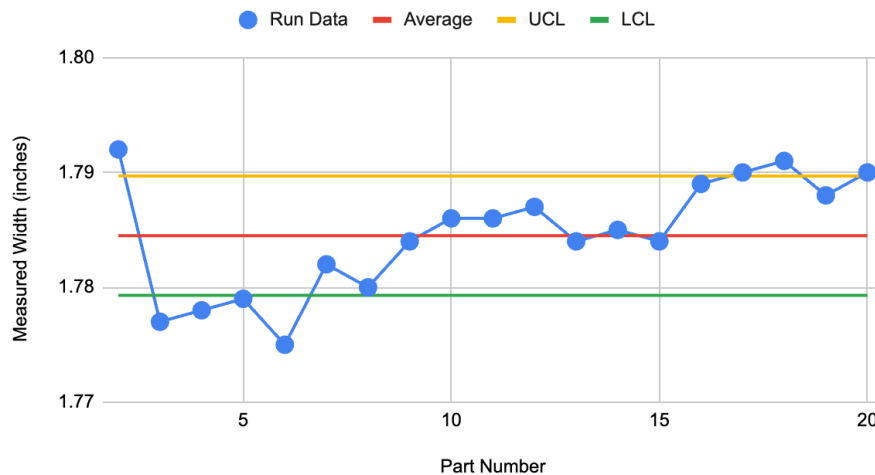


Figure 19. Run data of the bottom half widths

As seen in Figure 19, we created a run chart using our measurements from the 20 bottom half parts. While more parts fall within the UCL and LCL than outside of the control limits, similar to the top half, there are trends that we can identify that indicate that the process is unstable. Satisfying condition 1, eight points fall outside of the control limits. Given that points 2-6 are all below the LCL, condition 2 is also satisfied, and points 16-20 all fall outside of the secondary boundary, satisfying condition 3.

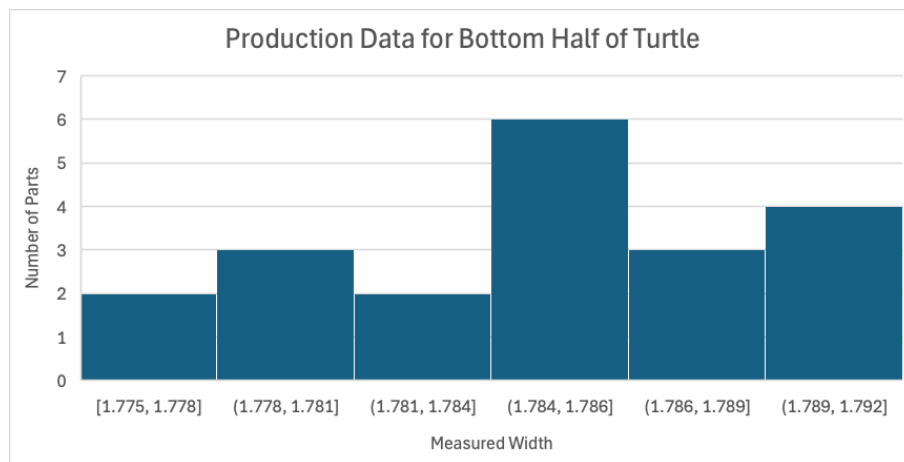


Figure 20. Histogram of measurement distribution of 20 bottom halves

From the histogram seen in Figure 20, we can see that our bottom parts also do not exhibit a normal distribution. Unlike the top half parts, only four out of the twenty bottom half parts fall within the specification limits.

Compared to the dimension we specified in CAD, none of the parts meet that value. This is likely due to shrinkage of the part during cooling. Deviations from the intended dimension can also be attributed to the errors we experienced during injection molding, including inconsistent nozzle temperature, pellet building, and deformation during ejection.

We also notice a larger distance from the specification limits for the bottom parts than the distance from the specification limits for the top parts of the turtle. In our design we included a draft angle and during the process we adjusted injection time and process procedures to achieve a better part quality and dimensional accuracy. We also injection molded all of the top parts first, and the increase in the cooling of the nozzle and barrel over time may be one reason for the specification limit discrepancy between the top and bottom parts. In terms of the mold, another reason could be a difference in the mold geometry that caused the mold for the bottom to be slightly thinner than intended. Overall, the greater deviation of the bottom parts is likely due to a combination of mold geometry differences, machining inaccuracies, and increased errors during the injection molding process.

VII. Conclusion

Overall, this project enabled us to work through the full process of designing, manufacturing, and injection-molding a product from initial concept and ideation to final production. Each phase of the project presented unique challenges that required problem-solving and design or process changes and refinement.

During the design phase, we ran into several problems modeling the parts. Complex geometries and small features were difficult to model and CAM due to unreliable and insufficient tool paths given the minimum tool size that we had access to. This led us to revisit the design and make necessary simplifications to better align with the CNC machining constraints and available tools. The simplification ultimately improved manufacturability and surface qualities of our final injection molded part.

During machining, issues with the initial press-fit feature dimensions highlighted the importance of accounting for tool limitations. The original small-diameter drill bit broke during operation, which forced us to redesign the pin and hole feature to accommodate a larger tool that would be less likely to break during CNC machining. While this modification allowed for successful machining, it resulted in a less precise interference fit and introduced larger variability in the final assembly of the turtle tops and bottoms.

The injection molding process also presented several challenges, including inconsistent nozzle temperature and material flow issues due to polymer pellet buildup. These factors led to short shots and variability in injection times. Through troubleshooting and process adjustments, we were able to produce parts with acceptable quality and surface finish. However, the increased variability in the process likely negatively affected the overall assembly performance.

For future consideration to improve our design, there are several modifications that can be explored. Incorporating a more accessible ejection feature could reduce some of the surface finish defects that occurred during removal of the part from the mold. Redesigning the press-fit features with additional tooling and/or with tighter tolerances and improved support during machining could also enhance the assembly accuracy and precision. Finally, exploring different runner designs could help ensure more consistent material flow and potentially reduce the amount of parts that were thrown out due to short shots.

Despite the challenges, the project gave us valuable hands-on experience with design for manufacturing, CNC machining, GD&T, and injection molding. The challenges we encountered highlighted the importance of adaptability, flexibility and careful, intelligent consideration throughout the manufacturing process.

VIII. Appendices

Appendix A: CAD Drawings of Parts

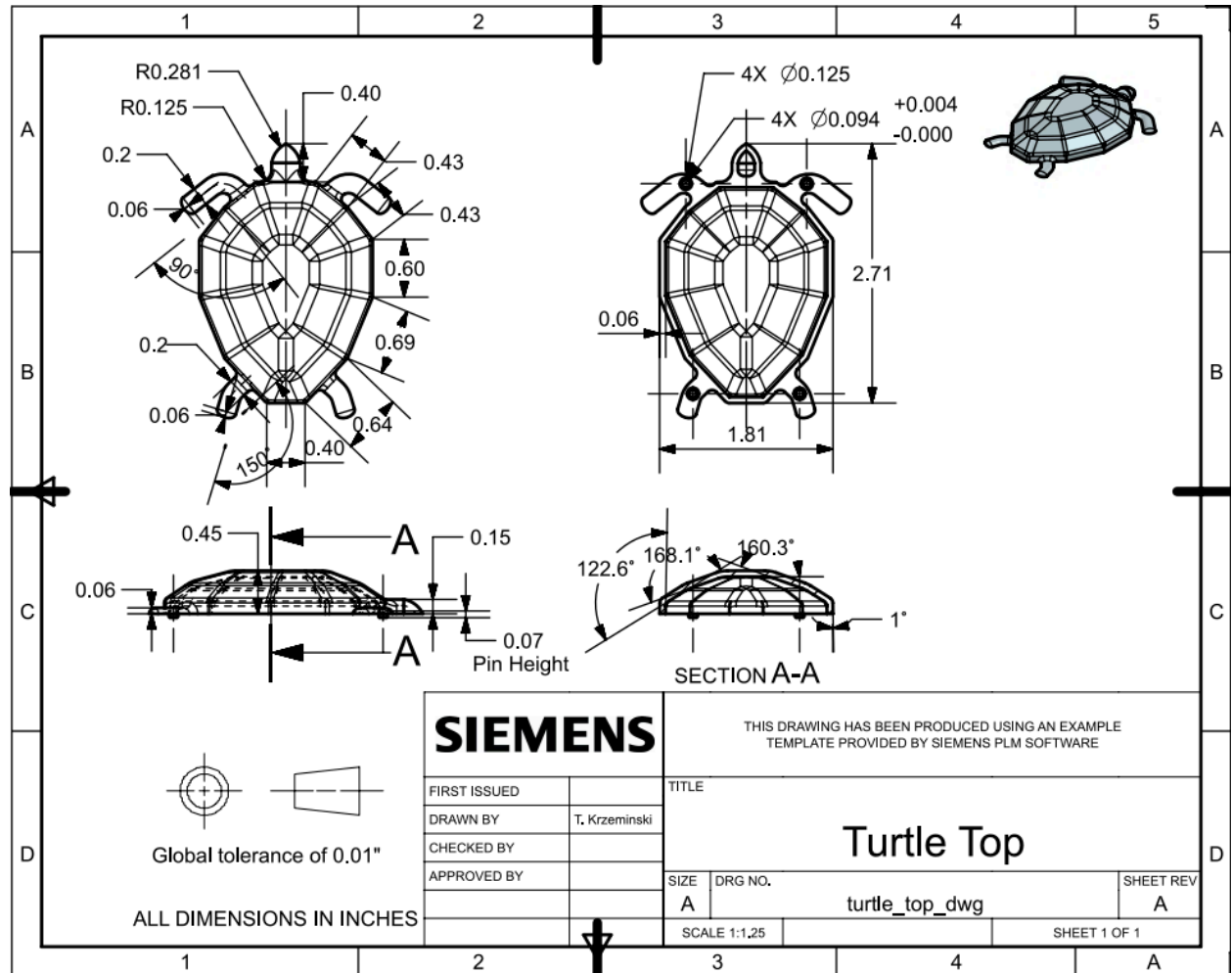


Figure 21: CAD Drawing of Turtle Top Shell

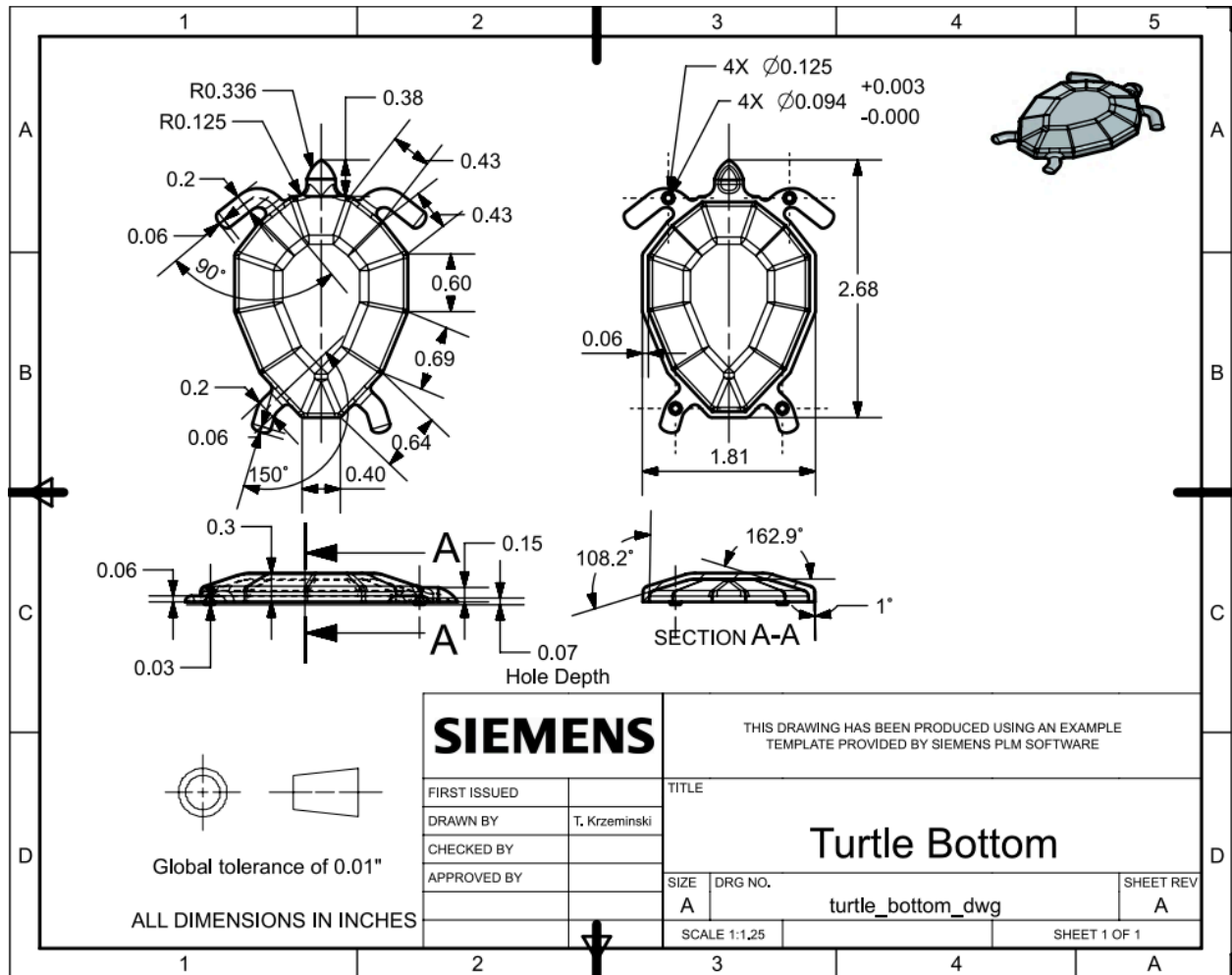


Figure 22: CAD Drawing of Turtle Bottom Shell

Appendix B: CAD Drawings of Molds

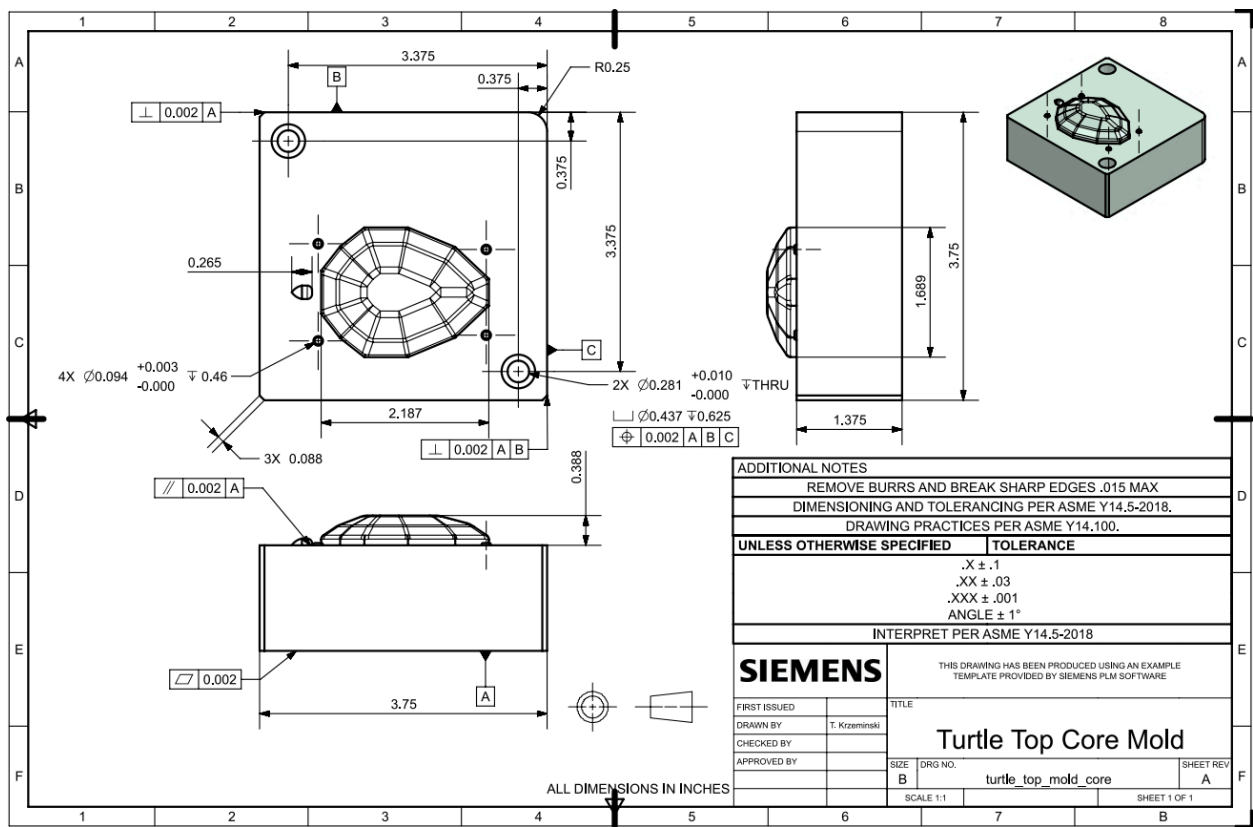


Figure 23: CAD Drawing of Turtle Top Core

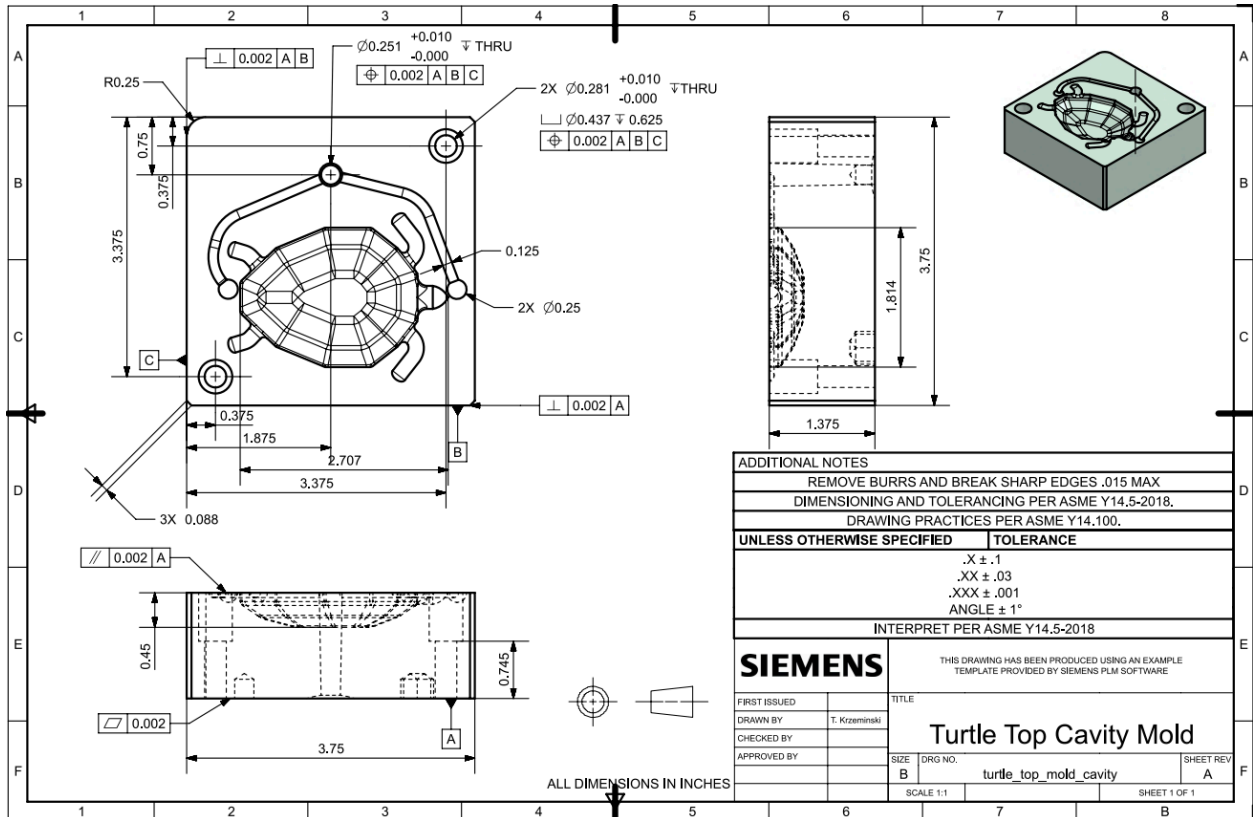


Figure 24: CAD Drawing of Turtle Top Cavity

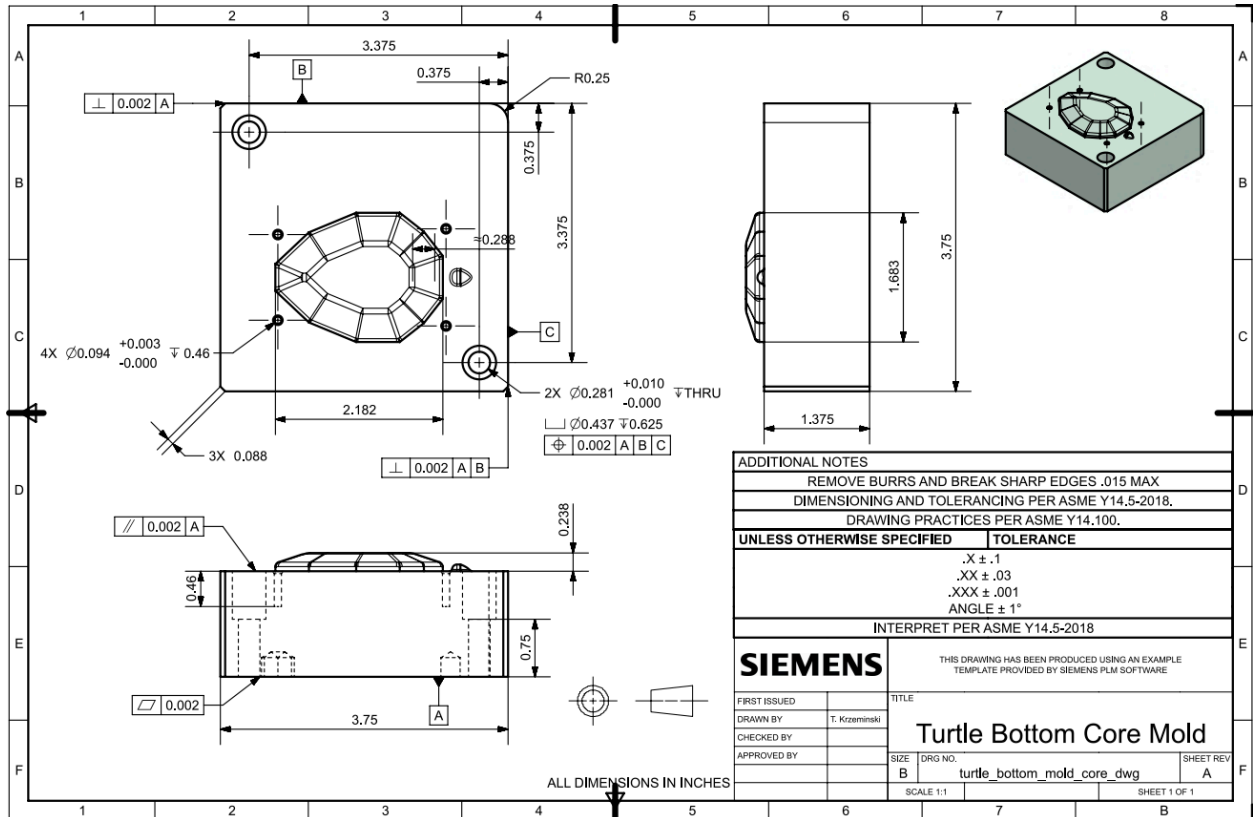


Figure 25: CAD Drawing of Turtle Bottom Core

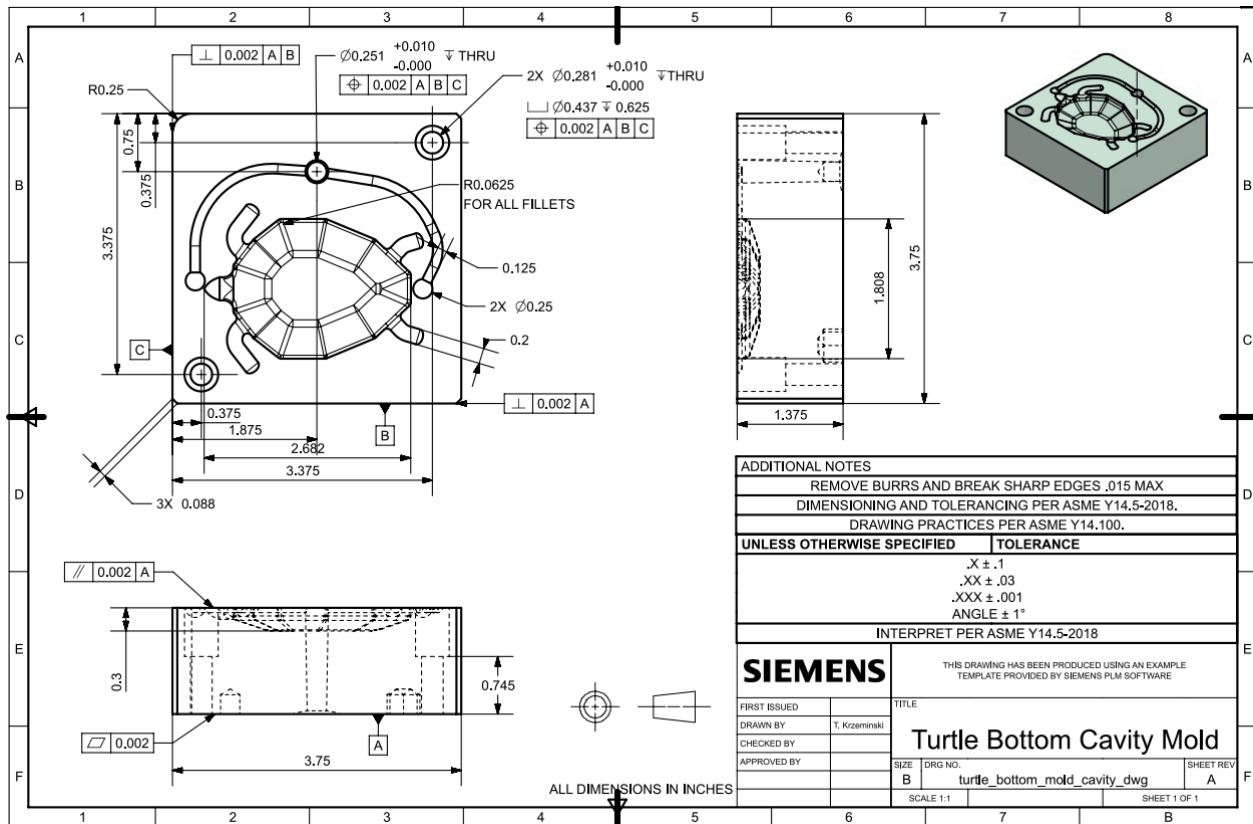


Figure 26: CAD Drawing of Turtle Bottom Cavity